Amir Mohammad Karimi-Mamaghan

Ph.D. student

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EECS, KTH Royal Institute of Technology, Stockholm, Sweden
Born on 1997-08-07 in Iran

EDUCATION

Present – August 2022 — **Ph.D. in Computer Science, KTH Royal Institute of Technology, Sweden**
- Division of Decision and Control Systems
- Supervisors: Bo Wahlberg, Stefan Bauer

May 2022 – September 2019 — **M.Sc. in Computer Engineering (Artificial Intelligence and Robotics), University of Tehran, Iran**
- Thesis: *Longevity Analysis of Cryptocurrency Addresses*
  I work on longevity (lifespan) of the addresses in popular cryptocurrencies (BTC, ETH, etc.) and try to fit a known distribution to this feature. I also try to find correlations between this feature and some other properties of the addresses.
- Laboratory: Data Analytics Lab, School of Electrical and Computer Engineering
- Advisor: Behnam Bahrak
- GPA: 18.17/20 (4/4)

July 2019 – September 2015 — **B.Sc. in Software Engineering, University of Tehran, Iran**
- Thesis: *Classification of Electrocardiogram Signals using Machine Learning & Neural Networks*
- Advisor: Siamak Mohammadi
- GPA: 18.5/20 (3.96/4)

RESEARCH INTERESTS

- Causal Inference
- Representation Learning
- Generative Models
- Combinatorial Optimization
- Operations Research
- Metaheuristics

ACADEMIC EXPERIENCE

Research Assistant:

- Research Assistant at Data Analytics Lab, University of Tehran, Iran, Feb 2020- May 2022.
  I did some projects to make Bitcoin transactions in P2PKH and P2SH modes and broadcast them to the blockchain. I also did some projects to make a simple application using Solidity, web3.js and Truffle framework in order to deeply understand smart contracts. Furthermore, I worked on bitcoin transaction data and analyzed some properties of the transaction graph and bitcoin addresses and tried to find correlations between them.

Research projects:

- Developing a mathematical model for the hub-and-spoke network under congestion, implementation of a meta-heuristic algorithm combined with a clustering technique for solving the problem, 2020.
- Design and implementation of a meta-heuristic combined with a reinforcement learning algorithm for solving a bi-objective stochastic resilient obnoxious p-median problem, 2021.
- Implementation of a Reinforcement Learning Algorithm for Mountain Car Problem, 2020.
  I implemented a Fuzzy model and a RBF model for a car stuck between mountains to help the car climb the mountains using Python and Gym.
- Design and Implementation of a Generative Adversarial Network (GAN) for Super Resolution Task, 2019.
  I implemented a GAN to increase the size of the images without diminishing the quality using PyTorch and Tensorflow.

- Implementation of a Hidden Markov Model - Gaussian Mixture Model (HMM-GMM) acoustic model for Automatic Speech Recognition Task, 2018.
  I implemented a HMM-GMM model to classify arabic digits audio dataset using PyTorch.
- Design and Implemenation of a custom CNN/RNN network for Question Answering Task, 2018.
  I developed a custom CNN/RNN network to automatically answer questions about an image using PyTorch.
- Analysis of Spotify songs popularity using Machine Learning Algorithms, 2020.
  I analyzed Spotify songs data (1920-2020) retrieved from Spotify API and used machine learning algorithms to predict popularity of the songs using Scikitlearn.

International Research Collaborations:

- Collaborating in an invited research project by the *European Journal of Operational Research* to do a review on the use of machine learning techniques in operations research for solving combinatorial optimization problems; **Collaboration with University of Lille and IMT Atlantique, France**, 2021, (Role: Bibliography Review, Technical Analysis and Discussion of Machine Learning techniques, Writing).
- Designing a hub-and-spoke network under congestion: A learning based metaheuristic; **Collaboration with Kedge Business School and IMT Atlantique, France**, 2020, (Role: Integration of Machine Learning into an Optimization Algorithm, Technical Development and Implementation).
- Developing a learning-based meta-heuristic for solving a bi-objective stochastic resilient obnoxious p-median problem; **Collaboration with Kedge Business School and IMT Atlantique, France**, ongoing, (Role: Implementation of a Learning-based Algorithm, Writing).

Teaching Assistant:

- **Deep Learning**: University of Tehran, Fall 2020
- **Database Design**: University of Tehran, Fall 2018
- **Discrete Mathematics**: University of Tehran, Fall 2018
- **Programming Languages and Compilers**: University of Tehran, Fall 2018
- **Algorithm Design**: University of Tehran, Fall 2017 & Fall 2018
- **Data Structures**: University of Tehran, Fall 2017 & Spring 2018
- **Engineering Probability and Statistics**: University of Tehran, Fall 2017
- **Web Development**: Allameh Helli High School (Exceptional Talents), August 2017-July 2020
- **C++ Programming**: Allameh Helli High School (Exceptional Talents), August 2018-July 2019

PUBLICATIONS

Journal articles (published)

- Karimi-Mamaghan, M., Mohammadi, M., Meyer, P., **Karimi-Mamaghan, A.** & Talbi, E. G., Machine Learning at the service of Meta-heuristics for solving Combinatorial Optimization Problems: A state-of-the-art. *European Journal of Operational Research*, 2021. (**JCR, IF** : 4.21) DOI
- Karimi-Mamaghan, M., Mohammadi, M., Pirayesh, **Karimi-Mamaghan, A.** & Irani, H., Hub-and-spoke network design under congestion: A learning based metaheuristic. *Transportation Research Part E: Logistics and Transportation Review*, 142, 102069, 2020. (**JCR, IF** : 4.69) DOI
- **Karimi-Mamaghan, A.**, Setayesh, A. & Bahrak, B. Analysis of Address Lifespans in Bitcoin and Ethereum, *12th International Conference on Computer and Knowledge Engineering (ICCKE)*, 2022, pp. 288-293. DOI

PROFESSIONAL EXPERIENCE

Data Engineer, Tapsell, November 2019-March 2020

- Tapsell is the leading company in the online advertising industry in Iran. During my experience at Tapsell, I implemented a Machine Learning model to map IP address of users to the location of each one. I also worked on Sauron, a module used for fraud detection and developed some heuristics to detect fraudulent users. I worked with Spring Boot, Kotlin, and Python and I got to work with big data processing tools such as Kafka streams, ElasticSearch, MongoDB and I got familiar with Apache Spark.

Software Engineer Intern, Rahnema College, July 2017-August 2017

- "Unagi", a location based social network application.
  In this project, I have focused on back-end development using Node.JS Express framework.

Honors and Awards

- **Gold Open Access award** for an *invited review* paper in the European Journal of Operational Research sponsored by EURO, 2021
- Ranked among the **top 5%** among **M.Sc.** students in Computer Engineering, University of Tehran, 2019-2021
- Awarded **full scholarship** for M.Sc. program in Computer Engineering at the University of Tehran, September 2019-May 2022
- Received **Straight Admission** without examination to M.Sc. in Computer Engineering Program at University of Tehran, September 2019
- Member of Iran's National Elites Foundation, 2015-Present
- **Awarded the FOE** (Faculty of Engineering) prize for three consecutive academic years, ranked **1st** among all **B.Sc.** students in Software Engineering, University of Tehran, 2015-2018
- Awarded **full scholarship** for B.Sc. program in Software Engineering at the University of Tehran, September 2015-May 2019
- Ranked 91 among 200'000 participants (**top 0.04%**) in the nationwide university entrance exam for undergraduate studies, (B.Sc. program), 2015

Main Courses

Machine learning	20/20	Artificial Intelligence	17.5/20
Data Analysis	17.5/20	Algorithm Design	20/20
Deep Learning	18.5/20	Bitcoin & Cryptocurrency Tech.	Coursera

Computer Skills

Programming	Python, R, C, C++, C#, Kotlin, Java, Matlab
Optimization	Gurobi, PulP, GAMS
Libraries and Frameworks	PyTorch, Tensorflow, Scikitlearn, Gym, Django, SpringBoot
Database	MySQL, NoSQLs(MongoDB, Cassandra, Redis), ElasticSearch
Tools	git, LaTeX, Docker, Kubernetes, KafkaStreams, Apache Spark

Language

English	●●●●○
TOEFL iBT	108/120 (Reading: 30, Listening: 27, Speaking: 23, Writing: 28)
GRE	Quantitative: 166, Verbal: 144, Writing: 4

References

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